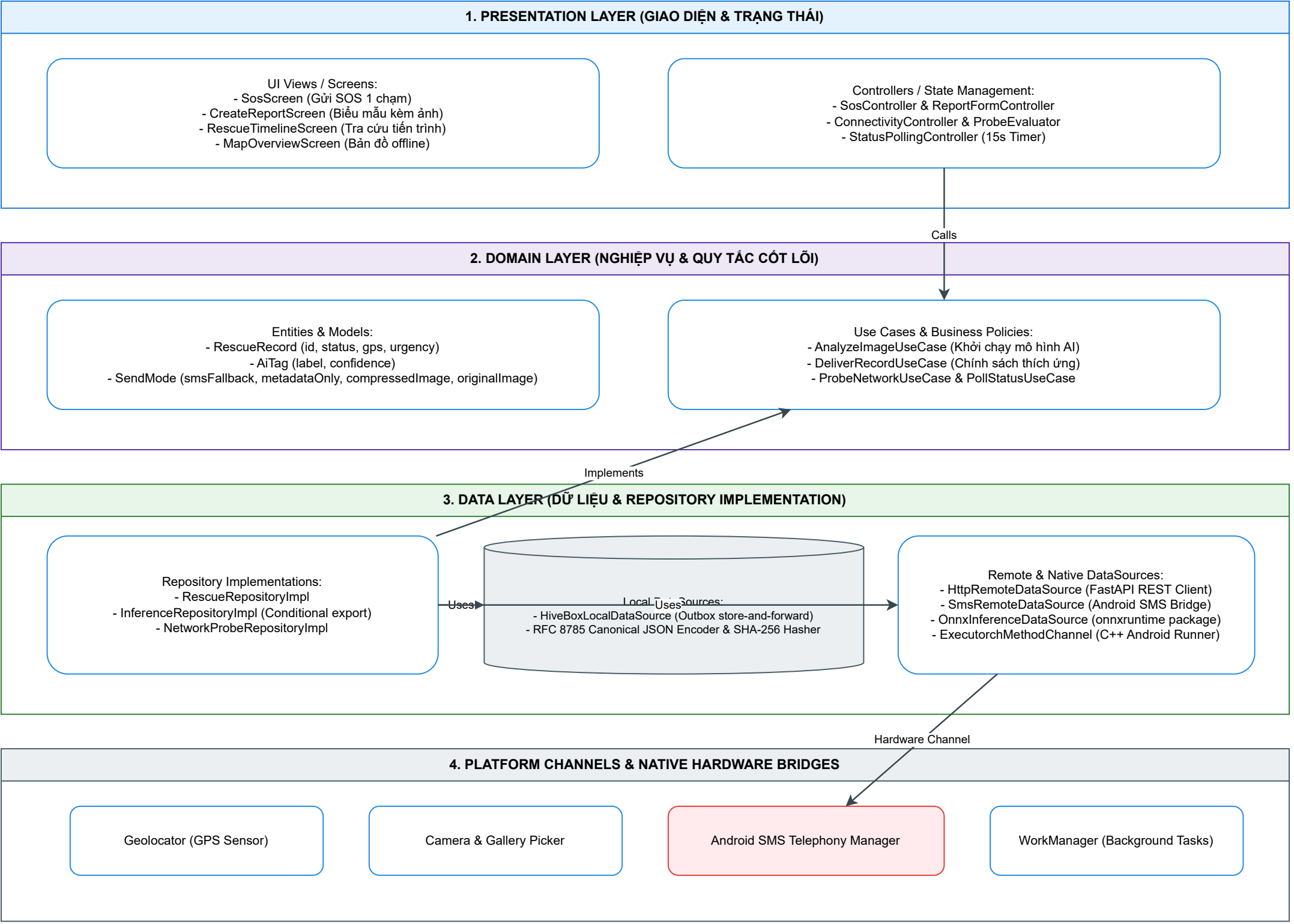

1. PRESENTATION LAYER (GIAO DIỆN & TRẠNG THÁI)
UI Views / Screens:
- SosScreen (Gửi SOS 1 chạm)
- CreateReportScreen (Biểu mẫu kèm ảnh)
- RescueTimelineScreen (Tra cứu tiến trình)
- MapOverviewScreen (Bản đồ offline)
Controllers / State Management:
- SosController & ReportFormController
- ConnectivityController & ProbeEvaluator
- StatusPollingController (15s Timer)
Calls
2. DOMAIN LAYER (NGHIỆP VỤ & QUY TẮC CỐT LÕI)
Entities & Models:
- RescueRecord (id, status, gps, urgency)
- AiTag (label, confidence)
- SendMode (smsFallback, metadataOnly, compressedImage, originalImage)
Use Cases & Business Policies:
- AnalyzeImageUseCase (Khởi chạy mô hình AI)
- DeliverRecordUseCase (Chính sách thích ứng)
- ProbeNetworkUseCase & PollStatusUseCase
Implements
3. DATA LAYER (DỮ LIỆU & REPOSITORY IMPLEMENTATION)
Repository Implementations:
- RescueRepositoryImpl
- InferenceRepositoryImpl (Conditional export)
- NetworkProbeRepositoryImpl
Uses
Uses
- HiveBoxLocalDataSource (Outbox store-and-forward)
- RFC 8785 Canonical JSON Encoder & SHA-256 Hasher
Remote & Native DataSources:
- HttpRemoteDataSource (FastAPI REST Client)
- SmsRemoteDataSource (Android SMS Bridge)
- OnnxInferenceDataSource (onnxruntime package)
- ExecutorchMethodChannel (C++ Android Runner)
Hardware Channel
4. PLATFORM CHANNELS & NATIVE HARDWARE BRIDGES
Geolocator (GPS Sensor)
Camera & Gallery Picker
Android SMS Telephony Manager
WorkManager (Background Tasks)